DESIGN AND FABRICATION OF PRESSURE VESSELS ACCORDING TO PD 5500

FWEL CONTRACT No : 1-14-16601/19

CONTRACT NO. : POGC-760-81-156

PROJECT : SOUTH PARS GAS FIELD DEVELOPMENT (PHASES 9&10)

COMPANY : NATIONAL IRANIAN OIL COMPANY

SITE : ASSALUYEH / SOUTH PARS GAS FIELD PERSIAN GULF, IRAN

REV.	DATE	DESCRIPTION	ORIG.	PRPD	CHKD	APP'D	CONTRACTOR's APP'D
8	08/12/04	AFC (Revised as marked)	LG	S.R.Kim	S.S.Park	H.S.Kim	S.C.Kim
7	31/08/04	Approved for Construction	LG	S.R.Kim	S.S.Park	H.S.Kim	S.C.Kim
6	15/07/04	Issued to Company	LG	S.R.Kim	S.S.Park	H.S.Kim	S.C.Kim
5	21/12/03	As per COMPANY's comments	LG/OIEC	S. R.Kim	H.C.Shin	K.S.Lee	
4	27/11/03	Issued for Feed Package	FWEL	S. Jacobs	S. Sim		
3	14/11/03	Issued for Feed Package	FWEL	S. Jacobs	S. Sim		S. Sim
2	12/11/03	Issued for Feed Package	FWEL	S. Jacobs	S. Sim		S. Sim
1	22/07/03	Issued for Company Review	FWEL	S. Jacobs	S. Sim		S. Sim
0	06/12/00	Issued for Basic Design Package	TECHNIP	V. Zarna	F. Durot		F. Durot

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– ADDENDUM
– TEP GENERAL SPECIFICATION N° SP-TCS. 212 Rev 0.

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GENERAL STATEMENT

The attached TOTAL EXPLORATION PRODUCTION (TEP) General Specification n° SP - TCS - 212 Rev. 0 is confirmed as the Project Specification, except as amended herein, and renumbered/reissued as specification n° RP 6340 S 999 0400 02 Rev. 7 for SOUTH PARS PHASES 9 & 10 Project.

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ADDENDUM

The following items refer to the attached TEP General Specification n° SP - TCS - 212 Rev. 0.
The clauses set out below shall modify or replace the existing clauses in the original specification as noted.

1. SCOPE

- Delete and replace the second sentence by :

- In this specification, high pressure vessels are those designed for a coincident pressure and temperature equal to or greater than class 600 rating as defined per ASME B16.5.

2. DEFINITION OF CONTRACTING PARTIES

• Delete and replace all the definitions :

GENERAL COMMENTS

- The following terms which are used in this general specification are deleted and replaced :

  – Engineering company by contractor : Shall mean the party which has entered under a Contract Agreement with POGC for the design, engineering, procurement and construction of the Onshore Facilities regarding the South Pars Field Development Project, Phases 9 &10.

  – TOTAL TEP : shall be read POGC or his nominated Representative.

  – Manufacturer : Shall mean any person, firm or company which manufacture or supply equipment or services for the performance of any item of the work.

  – Work : Shall mean works and / or services entrusted by contractor under the contract with POGC.

3.1 Basic code

- Delete and replace the paragraph by :

– The Basic Code is the British Standard Institute Published Document PD5500 : Specification for unfired fusion welded pressure vessels. This Code is referred to herein as "the Code".

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3.3 Other Codes and Standards

- Delete and replace :

– ANSI B58.1 by UBC Uniform Building Code, Volume II. Wind and Earthquake.

– MSS SP 44 by ASME B 16.47, A Series.

- Add :

– ASME B16.9 : Factory - Made Wrought Steel Butt Welding Fittings

– ASME B36.10 M : Welded and Seamless Wrought Steel Pipe

– WRC Welding Research Council Bulletin 107.

3.4 Total specifications :

- Delete and replace :

– SP-TCS-611 by RP 6340 S 999 6300 01

3.5 Total standard drawings :

- Replace by the list given in:

– RP 6340 S 999 0400 05 Appendix 1.

4.1.3 Design Pressure (DP)

- Delete the fourth paragraph of section a) and replace by:

– The maximum discharge pressure at zero flow of the pump or compressor feeding the vessel (i.e. head at zero flow plus design suction pressure).

- Add to end of section a):

– Internal static head shall be added to the design pressure to determine the thickness of any specific parts of the vessel.

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4.4 Corrosion Allowance

- Delete section a) and replace by:

– a) Additional thickness shall be provided for the corrosion allowance specified in the individual data sheet. In the absence of a specified value the minimum corrosion allowance shall be 3mm for carbon steel and 0 for stainless steel, high alloy or clad steel.

4.5.1 Basic code

- Delete entirely three paragraphs and replace by :

– The Basic Code is UBC - Uniform Building Code Volume II, for wind. Refer to project specification Civil Design Criteria RP-6340S-999-1782-001 for specific design details.

4.6.1 Basic code

- Delete the paragraph and replace by :

– The Basic Code is UBC - Uniform Building Code Volume II, for Earthquake. Refer to project specification Civil Design Criteria RP-6340S-999-1782-001 for specific design details.

Add new sections

4.7 Steam out

Where applicable, vessels shall be designed for steam out at 5°C / full vacuum or 150°C/ATM. Except units 104/105/106/111/147/148/149 and 150. If applicable, necessity of steam out condition shall be specified on the individual process data sheet according to process requirement.

4.9 Unit Loads

For design purposes, unit loads of vessel attachments are as follows:

– Platform: 120 kg/m^2 (including hand rail and support members)

– Ladder: 35 kg/m

– Insulation: 230 kg/m^3

– Fire proofing: 2200 kg/m^3

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5.2 Combinations of loads

- Add: Allowable stresses shall be in accordance with PD 5500 Annex A.

5.3 Local loads :

- Delete and replace the second sentence by :

– The methods for computing such stress shall comply with PD 5500 appendix G. Allowable stress combination for this purpose shall comply with appendix A of the code. In the absence of specified nozzle loadings, loads from RP-6340-S-999-0400-003 Rev.0 included in The Basic Design Package shall be used.

5.7.3 Skirts of Vertical Vessels :

- Add :

- Minimum skirt thickness shall be 10mm or 1/3 of the lowest course shell thickness, which is greater one. For conical skirts maximum cone angle shall be 15 degrees.

5.7.3 Nozzles :

- Delete and replace **5.7.3 Nozzles** by **5.7.4 Nozzles**

- Delete and replace :

- The standard Total No. PS-TCS-211 by:

The standard for minimum nozzle thickness, PS-TCS-211, contained in RP-6340-S-999-0400-005.

- Add the sentence :

- Nozzle necks shall meet the requirements of the paragraph 3.5.4.7. of PD 5500.

6. MATERIALS

6.1 – Selection of materials

- Delete and replace BS 5500 by PD 5500

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- Add:

– Any plate over 40mm nominal thickness and any plate which is to be cold worked to form a dished head, shall be supplied in the normalised condition unless the relevant material specification states that such treatment is detrimental to the properties of the plate.

6.3**. Weldability of C-Mn steels**

- Delete and replace :

– *Carbon Content (C) 0.23 by : 0.22*

– Carbon Equivalent (CE) 0.42 by : 0.43

6.4 **Flange Gaskets and Contact Surfaces:**

- Delete and replace Asbestos by Non-asbestos.

6.5.1 External bolting

- Delete and replace the second sentence of the first paragraph :

– They shall be protected against corrosion in accordance with RP 6340 S 999 6600 001 paragraph 4.4.

7.3 Welded joint layout

- Delete and replace section d) by :

– d) No attachment weld edge shall be closer to the nearest weld edge than 1.5 times the wall thickness or 40mm whichever is the greater.

7.7 Shell flanges sub-paragraph a

- Delete and replace :

– MSS SP 44 by : ASME B16.47 A Series.

- Delete and replace :"Alternatively : ASME …." by :

– Outside the limits of the above standards, body flanges shall be designed according to the Code.

Shell flanges sub-paragraph b

- Delete and replace the sentence by :

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– The flanges shall be designed in accordance with the Code.

7.8.1 Vertical vessels - sub-paragraphs a) and b)

▪a) Delete and replace :

– first sentence by: All vertical vessels shall be designed to be free standing and self supporting on a cylindrical or conical skirt in accordance with the standard detail 6340S 000 STC 000400 301 & 304 shown in RP-6340-S-999-0400-005.

▪b) Delete and replace :

– TOTAL standard PS-TCS-209 by: standard detail 6340S-L-C021 shown in RP-6340-S-999-0400-005.

▪c) Delete and replace :

– No brackets, lugs, nor other type of support shall be used, except using for special vertical case only which should be installed on Platform or Steel structure, not on ground directly.

7.8.2 Horizontal vessels

▪ Delete and replace :

– PS TCS 207 by: standard detail PS TCS 207 shown in RP-6340S-999-0400-005

7.9.1 Reference standard - sub-paragraph a

▪ Replace MSS SP 44 by :

– ASME B16.47 A Series.

7.9.1 Reference standard - sub-paragraph b

▪ Add : Rolled plate nozzle neck, which are only acceptable for over than 16” nominal size and performed successfully full Radiographic Examination (RT), shall be of the same material as specified for the vessel shell / head to which they are attached. Reinforcing pads shall be of the same material as the shell or head to which they are attached.

7.9.2 Sizes

▪ Delete and replace TOTAL standard PS-TCS-212 by: standard PS-TCS-212 shown in RP-6340-S-999-0400-005

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7.9.3 Reinforcement

- Add:

Self-reinforcing forged nozzles shall be used when required by Code or where any of the following conditions are met:-

1) Design temperature exceeds 450°C.
2) Design Pressure exceeds 100 barg.
3) Design temperature exceeds 350°C and design pressure exceeds 30 barg.
4) Plate thickness exceeds 50mm.

7.9.6. Manholes

Delete and replace section a) by:

a) For vessel diameter < 1000mm flanged vessel will be considered if equipment contains internals, otherwise 18" diameter manhole shall be provided. For vessel diameter ≥ 1000mm manholes shall be 20" diameter in nontoxic service and 24" in toxic service.

- Delete and replace section b) first sentence by:

b) Manholes shall conform to standards PS-TCS-214, PS-TCS-216 and 6340S-L-C008 shown in RP-6340-S-999-0400-005.

7.9.7. Connection attachment to shell

- Delete and replace section a) by:

a) Typical details are as per PS-TCS-213 shown in RP-6340-S-999-0400-005.

7.10 Other attachments to vessel walls sub-paragraph b

- Delete and replace :

c) Fixed internals may include rings, bars, brackets and clips to support trays and down-comers, piping, gratings and mesh, distributors, etc… as well as baffles weirs, impingement and wear plates, ladder rungs, etc. Typical details for some of the above are contained in RP-6340-S-999-400-005.

d)

7.11. Lifting Devices

- Delete and replace TOTAL Standard PS-TCS-204 by : standard 6340S-L-C022 shown in RP-6340-S-999-0400-005.

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8.4 Welding

- Delete and replace the first sentence by :

– Welding shall comply with the Basic Code PD 5500 and RP 6340 S 999 6300 001, and the following additional requirements :

- Add to sub-paragraph 8.4.1 d) :

– For vessel wall thickness 12 mm and below, back-gouging may be performed by grinding only provided that all the relevant welds are 100 %, X-ray inspected.

8.5.2 Specific requirements - sub-paragraph a

- Delete and replace NACE 01-75 by NACE 0175

10.4 Material verification, inspection and testing

10.4.3 Material test reports

- Delete and replace DIN 3.1B by EN 10024 3.1.B
- Delete and replace the last sentence by :

- Material certificates for non-pressure parts shall conform to EN 10024 2.2

10.4.5 General

- Delete and replace in the second sentence :

ASME Code section V and SA 435 by :

– BS EN 10160 classes S2, E2

10.5 Welding inspection

10.5.1 Visual and non destructive examinations

- Delete and replace :

– SP-TCS-611 by RP 6340 S 999 6300 01

10.5.1 h)

- Delete and replace :

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Radiographic inspection must be done by either x-rays or gamma rays due to thickness. However, radiography shall be performed using gamma rays only if AGFA GEVAERT D4, KODAK M or FUJI 1X80 film is used.

10.5.1 i) Qualification of NDT personnel

- Replace SNT - TC -1A Level I or Level II by :

– European norm. EN 473 entitled “Qualification and Certification of NDT Personnel” and/or para.5.6.1 of PD 5500.

10.5.3 Dimensional Check

- Delete and replace :

– TOTAL standard No. PS-TCS-201 by: standard PS-TCS-201 shown in RP 6340 S 999 0400 005.

11.3. Name plate

- Delete and replace :

– TOTAL standard No. PS-TCS-223 by : standard 6340S 000 STC 000 400 104 shown in RP 6340 S 999 0400 005.

- Add the following sections:

11.4 Transportation & Installation

Transportation and installation requirements shall be reviewed by the manufacturer who will check the integrity of the vessels and provide any additional stiffening required prior to shipment.

11.5 Dimensions

Where dimensions are specified in imperial units, manufacturer shall fabricate to the nearest metric equivalent, (excluding nominal pipe diameter).

11.6 Tailing Lugs

Tailing lugs, when required, must be located on the skirt and not on the vessel wall. They shall be attached to the skirt base ring ensuring no interference with vessel hold down bolts.

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TOTAL EXPLORATION PRODUCTION

GENERAL SPECIFICATION

PIPING - VESSELS - WELDING

SP - TCS - 212

DESIGN AND FABRICATION

OF PRESSURE VESSELS

ACCORDING TO BS 5500

Rev. 0	January 93	First issue

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1 - SCOPE

This General Specification covers TOTAL practices and minimum requirements for the engineering design, selection of materials, fabrication, inspection and testing of fusion welded unfired pressure vessels conforming in definition to the Code mentioned in 3.1.

In this specification, high pressure vessels are those designed for a coincident pressure and temperature equal to or greater than the pressure-temperature ratings shown in Table G-600 of ASME B16.5 : Pipe Flanges and Flanged Fittings (i.e. equal to or higher than 600 Lbs rating).

2 - DEFINITION OF CONTRACTING PARTIES

OWNER	The legal Owner of what is built.
COMPANY	The party which signs the contracts and orders on behalf of the OWNER.
TOTAL	TOTAL acts as the COMPANY representative and performs the role of MANAGER.
ENGINEERING COMPANY	This term designates the engineering companies which work under the supervision of TOTAL and assume the function of Engineer. In all cases a contract exists between the COMPANY and the ENGINEERING COMPANY specifying the missions entrusted to the latter.
MANUFACTURER	The Company responsible for fabrication of pressure vessel(s), either in its facilities or on site.
INSPECTION AGENCY	This term designates the inspection bodies which work under the supervision of TOTAL. In some cases, TOTAL may also act as the INSPECTION AGENCY.

3 - CONTRACTUAL REFERENCE DOCUMENTS

Pressure vessels covered by this specification shall be designed, engineered, and built in accordance with the latest revisions issued at the date of the call for tender of the following documents listed hereafter.

Any divergence between any of the Contractual Reference Documents, or between this specification and any other Contractual Reference Document, shall be reported to TOTAL for decision. In such a case, and unless otherwise agreed or decided by TOTAL, it is understood that the more stringent requirement shall apply.

Exceptions to, or deviations from this specification are not permitted unless previously accepted in writing by TOTAL. For this purpose, MANUFACTURER's requests for substitution or change of any kind shall be complete with all pertinent information required for TOTAL assessment.

3.1 - Basic code

The basic code is the British standard BS 5500 : Specification for unfired fusion welded pressure vessels.

3.2 - Local regulations

If the country of installation has its own regulation, it must be applied wherever more stringent than the Basic Code.

Any Construction Code or Government Regulations which may be listed in the Project Specifications, in the requisition or in the Purchase Order, if any, shall have precedence except where this Specification is more stringent.

3.3 - Other codes and standards

- ASME	B 1-1	:	Unified inch screw threads.
	B 16-5	:	Steel pipe flanges and flanged fittings.
	B 16-11	:	Forged steel fittings, socket welding and threaded.
	B 16-20	:	Ring joint jackets and grooves for steel pipe flanges.
	B 16-21	:	Non-metallic gaskets for pipe flanges.
	B 16-25	:	Butt-welding ends.
	B 16-30	:	Unfired pressure vessel flange dimensions.
	B 46-1	:	Surface texture.
	B 58-1	:	Building code requirements for minimum design loads in building and other structures.
- MSS	SP 6	:	Standard finishes for contact faces of pipe flanges.
	SP 44	:	Steel pipe line flanges.

3.4 - TOTAL specifications

- SP-TCS-611 : Welding of pressure containing piping and equipment.

3.5 - TOTAL standard drawings

- PS-TCS-201 : Tolerances of pressure vessels.
- PS-TCS-202 : Horizontal tanks, heads with large knuckle radius.
- PS-TCS-203 : Horizontal tanks, elliptical heads.
- PS-TCS-204 : Lifting lugs.
- PS-TCS-205 : Anchor lugs and base rings of vertical vessels.
- PS-TCS-206 : Saddles for horizontal containers, diameter 350 to 500 mm.
- PS-TCS-207 : Saddles for horizontal tanks.
- PS-TCS-208 : Standard equipment of column skirts.
- PS-TCS-209 : Legs for small vessels.
- PS-TCS-210 : Nozzle designation.
- PS-TCS-211 : Minimum nozzle thicknesses.
- PS-TCS-212 : Nozzles less than 2" in diameter.
- PS-TCS-213 : Protruding length and bolt holes of nozzles.
- PS-TCS-214 : Vertical manhole with davit (WN flange).
- PS-TCS-215 : Vertical manhole with davit (SO flange).
- PS-TCS-216 : Horizontal manhole with davit (WN flange).
- PS-TCS-217 : Horizontal manhole with davit (SO flange).
- PS-TCS-218 : Hinged manhole (WN flange).
- PS-TCS-219 : Hinged manhole (SO flange).
- PS-TCS-220 : Handles and rungs for manholes.
- PS-TCS-221 : Anti-vortex.
- PS-TCS-222 : Deflector on steam inlet.
- PS-TCS-223 : Plates for vessels.
- PS-TCS-224 : Earthing plate.
- PS-TCS-225 : Monorail brackets.
- PS-TCS-226 : Piping gussets.
- PS-TCS-227 : Platform lugs.
- PS-TCS-228 : Ladder lugs.
- PS-TCS-229 : Fire-proofing supports.
- PS-TCS-230 : Insulation supports on vertical vessels.
- PS-TCS-231 : Insulation supports on horizontal vessels.
- PS-TCS-232 : Welding of cladded vessels.
- PS-TCS-233 : Coating of nozzles.
- PS-TCS-234 : Weld deposits on RTJ seats.

3.6 - Individual specifications

- Data sheets and drawings
- Project specifications and standards
- Requisition, notes etc.

3.7 - MANUFACTURER's responsibility

This specification together with the individual specifications shall not be construed as releasing the MANUFACTURER from his responsibility, fulfillment of his duties, guarantee and/or liabilities, nor does TOTAL, by any acceptance or approval, assume any responsibility for the conformance of the vessels to the requirements of the CODE and/or of the Governing Authority.

4 - DESIGN DATA

4.1 - Pressures

All pressures mentioned are gauge pressures except in the case of vacuum vessels.

4.1.1 - Service pressure (SP)

This is the pressure applied to the vessel during its normal operation.

4.1.2 - Maximum service pressure (MSP)

This is a temporary peak pressure that may have a variety of causes, such as : adjustment during starting up, increase in SP through accidental operation, momentary overpressure at a certain stage of operation ; it is therefore the maximum pressure anticipated during the complete operating cycle of the vessel under the most severe conditions.

4.1.3 - Design pressure (DP)

a) Operation above atmospheric pressure

The design pressure shall be taken as the largest of the following values :

- the maximum service pressure (MSP) with a safety margin according to the following table :

MSP (barg)	Design pressure (barg)
0 - 10	MSP + 1 bar
10 - 35	MSP + 10%
35 - 70	MSP + 3.5 bar
> 70	MSP + 5%

- a minimum of 2 barg,
- the minimum discharge pressure at zero flow of the pump or compressor feeding the vessel (i.e. head at zero flow plus design suction pressure),
- the vapor pressure at design temperature of the fluid contained, except where safety relief valves are designed to take into account the corresponding increase of pressure.

b) Operation under partial vacuum

If the internal pressure is 0.35 bar or less, the vessel shall be designed for full vacuum.

In all other cases, the vessel shall be designed for the maximum external pressure plus a margin of 0.15 bar.

4.2 - Temperatures

4.2.1 - Service temperature (ST)

This is the temperature to which the vessel is exposed during its normal operation.

4.2.2 - Maximum service temperature (MST)

This is a temporary peak temperature reached :

- systematically during an operating cycle,
- occasionally :
  - . due to foreseeable operating error,
  - . due to the failure of some safety device,
  - . due to unit shut down.

If an operating error or the failure of a safety device should result in a large foreseeable increase in the ST, and if this increase could affect a limited zone of the vessel, this fact should be taken into account and therefore stated. It is therefore possible to specify several MSTs for a given vessel.

4.2.3 - Minimum service temperature (mST)

This is the lowest temperature possible during normal operation (this temperature may be caused by internal fluids or by the ambient temperature). The low temperatures reached during occasional operation (such as depressurization) shall be mentioned on the process data sheets, stating whether they should be taken into account. The basic code indicates to what extent these temperatures shall be taken into account for material selection.

4.2.4 - Design temperatures (DT)

These temperatures are taken into account for material selection. They are determined as follows :

- DT = MST + 15°C if MST is equal to or greater than ambient temperature,
- DT = mST - 5°C if mST is below ambient temperature.

However, subject to TOTAL prior approval on a case by case basis, these rules may be waived for material selection where economics would strongly indicate to stick to DT = MST or DT = mST as applicable.

4.3 - Combination units

The partition between two (2) independent chambers of a combination unit shall be designed to withstand the design pressure and temperature of any chamber whilst the adjacent chamber is under atmospheric pressure and temperature.

Unless otherwise authorized in writing by TOTAL, design based on the differential pressure is not permitted.

4.4 - Corrosion allowance

a) Additional thickness shall be provided for the corrosion allowance specified in the individual data sheet.

b) Corrosion allowance for nozzles and manholes shall be at least equal to that specified for the vessel shell. No corrosion allowance is required on gasket face of flanges.

c) For fixed internals, corrosion allowance shall be added to both sides of the components.

4.5 - Wind

4.5.1 - Basic code

The basic code is ASME A-58-1 ; "Building code requirements for minimum design loads in buildings and other structures", which must be applied as design method.

In addition to the static calculations, calculations taking account of dynamic effects must be carried out for all vertical vessels, such as vertical drums, columns, stacks, etc..., in application of Appendix A 6.3.4.1 of ASME A-58-1, when the slenderness ratio is greater than 5.

However, in the case of very slender or special vessels, TOTAL and/or the ENGINEERING COMPANY may require that the MANUFACTURER furnish calculations under dynamic conditions based on the references of the code or on other methods.

4.5.2 - Design data

Wind velocity shall be in accordance with the site meteorological conditions.

4.6 - Earthquake

4.6.1 - Basic code

The basic code is ASME A 58-1 "Building code requirements for minimum design loads in buildings and other structures". It must be applied as design method.

4.6.2 - Design data

The individual specifications give, as appropriate, the numerical data necessary, in particular coefficient "Z" applicable to the site of installation.

5 - DESIGN RULES

5.1 - Special design cases

Cases and features not fully covered by the CODE shall be the subject of agreement in writing between the MANUFACTURER and TOTAL.

5.2 - Combinations of loads

All vessels and their supports, including anchor bolts, shall be designed for the most adverse condition to which they may be subjected.

For that purpose, in addition to sizing the various components for the most severe condition of coincident design pressure and temperature, the MANUFACTURER shall determine the stress level in the shell and support resulting from all the combinations of static and dynamic loads which may occur at any time during the vessel's life.

The extent and methods of that investigation and the appropriate data to be used in calculations shall be agreed about as early as possible (and preferably before contract award) between TOTAL and the MANUFACTURER.

As a minimum, the MANUFACTURER shall analyze the structural stability of the vessel and its support for the combinations of loading and conditions given in table 5.2.

Table 5.2

Combination of loadings

	CONDITION		
LOADING	AS INSTALLED	OPERATING	HYDROSTATIC TESTING (1)
Dead load	(2)	(3)	(3)
Live load	-	(4)	(5)
Internal and external pressure	Atmospheric	(6)	(7)
Temperature	Atmospheric	(8)	Atmospheric
Wind	(9)	(9)	-
Earthquake	(10)	(10)	-

Notes : (1) In operating position.

(2) "Lifting weight" of the vessel, empty, new (non corroded), including all fixed attachments, excluding loose internals and appurtenances, excluding insulation.

(3) "Operating weight" of the vessel, corroded, including any permanent equipment attached to the vessel, e.g. internals, platforms, ladders, piping, rotating equipment and drive, insulation, fireproofing.

(4) Weight of normal liquid content.

(5) Weight of water.

(6) Design pressure, including static head, vessel full of water.

(7) Test pressure, including static head, vessel full of water.

(8) Design temperature.

(9) See 4.5.
Shall not be combined with seismic load.

(10) As specified in the individual specification.
Shall not be combined with wind load.

5.3 - Local loads

The MANUFACTURER shall check the localized stresses and deformations in the vessel wall due to concentrated loads caused by external and/or internal structural attachment, piping reactions, supported equipment, etc.

The methods for computing such stresses shall previously be submitted to TOTAL acceptance (typical methods of calculations are given in the British Code BS 5500 : Specification for Unfired fusion welded pressure vessels).

5.4 - Maximum allowable stress

The MANUFACTURER shall verify that, in any combination of loadings, except the testing condition, and in any part of the vessel and support, the computed stresses do not exceed the values prescribed in the CODE.

Under the condition of the hydrostatic test, the stress shall not exceed 95 percent of the minimum specified yield strength of the material at testing temperature. The efficiency of full penetration butt-weld joints shall be taken as 1.

5.5 - Maximum allowable working pressure

The maximum allowable working pressure shall be limited by the shell, heads or flanges only, not by minor components such as nozzle neck, reinforcing pad, etc.

This requirement does not apply to vessels governed by 5.7.

5.6 - Deflection

Lateral deflection of vertical vessels under the effect of dynamic loading shall not exceed the following :

- Columns with internal trays : $\frac{H}{300}$

- Other vessels : $\frac{H}{100}$

Where H = Total height of the vessel and support.

5.7 - Minimum thicknesses

5.7.1 - Shell and heads

The minimum thickness of shell and heads shall not be less than :

$$t = \frac{D}{800} + 3 + C$$

t = Minimum thickness in mm (rounded up to next higher mm)

D = Outside diameter in mm

C : Corrosion allowance in mm

This requirement applies to :

a) Walls consisting of a single material (i.e. carbon, low alloy, or stainless steel, etc...).

b) Compound walls made of carbon steel (or low alloy steel) with layers of stainless steel. In this case, the thickness of stainless steel may be included in the minimum thickness only in the case of sheets plated by rolling or cladding, to the exclusion of stainless steel linings.

5.7.2 - Internals

The A columns of table 5.7 hereafter give the minimum thicknesses for internals, including the corrosion allowance, as a function of the corrosion allowance specified for the vessel.

The B columns give the corrosion margin in mm included in the A column, applied as indicated by note (1) to only one side or to both sides, according to whether or not the item is removable.

How to read table 5.7 :

a) Select the column according to the corrosion allowance required for the shell.

b) If the component is not subjected to loads requiring larger dimensions, its minimum dimensions will be selected from column A.

c) Otherwise, these dimensions shall be calculated, and the corrosion allowance indicated in column B shall be added to one or both sides according to the indication of note (1) of the table.

5.7.3 - Skirts of vertical vessels

Skirt thickness shall be a minimum of one third of the main shell thickness. In case of pressure vessels made of several sections, the skirt minimum thickness shall be taken as one third of the largest shell thickness value. (However, larger values of skirt thickness may still be required as a result of code calculations).

Table 5.7

Internals

Corrosion allowance required on the main wall of the vessel (mm)		0		1.5		3		5		6	
		A	B	A	B	A	B	A	B	A	B
Carbon-steel trays (weirs) Stainless-steel downcomers (11/13 cr)		3 2 US gauge= 14	0 0	3 2 US gauge= 14	- -	3.5 2 US gauge= 14	- -	6 2.7 US gauge= 12	- -	6 3.5 US gauge= 10	- -
Removable partitions, baffles, basins (carbons steel etc)		3	0	4	1.5 (1)	6	3 (1)	7	5 (1)	8	6 (1)
Permanent partitions, baffles, stacks, basins, etc... or compartment separators (carbon steel)		5	0	6	1.5	8	3	12	5	14	6
Gratings and grids (carbon steel) Minimum thickness of rods		3	0	4	1.5 (1)	6	3 (1)	7	5 (1)	8	6 (1)
Internal bolts, dia. in mm (carbon or stainless steel)		10	-	10	-	10	-	10	-	12	-
Unpressurized internal tubes (carbon steel)	diameter ≤ 2"	STD	0	STD	1.5 (1)	XS	3 (1)	XXS	5 (1)	XXS	6 (1)
	diameter > 2"	STD	0	STD	1.5 (1)	STD	3 (1)	XS	5 (1)	XS	6 (1)
Support beams (carbon steel)	Main	-	0	-	1.5	-	3	-	5	-	6
	Secondary	-	0	-	1.5 (1)	-	3 (1)	-	5 (1)	-	6 (1)
Support rings of trays, downcomers, demisters, etc... Carbon steel Stainless steel		10 6	0	10 6	1.5	10 7	3	13 7	5	14 8	6

Note (1) : Corrosion allowance on only one side.

Column A : Minimum dimension is thickness or diameter as appropriate.

Column B : Corrosion allowances (included in column A).

5.7.3 - Nozzles

The standard TOTAL n° PS-TCS-211 gives, for each diameter and according to the corrosion allowance, the minimum thickness including the fabrication tolerance of minus 12.5%. It states in particular that the thickness of pipes may be limited to the thicknesses of the walls to which they are welded, except in the special case of allowances required by abrasion, for example. The MANUFACTURER may select greater thicknesses in so far as the resulting inside diameters are large enough for whatever must be run through the nozzles.

6 - MATERIALS

6.1 - Selection of materials

According to the design data and geometrical characteristics of the vessels, the plate materials shall be selected among those listed in the basic code used for the design and construction of the vessel.

Other materials including forged materials may conform to either the BS 5500 or the ASME codes listed materials.

The selection of materials as a function of the design data must be made in conformity with Table II. TOTAL and/or the ENGINEERING COMPANY specifies the materials for each vessel which may be different from the materials of Table II in certain special cases arising from the nature of the fluids or from corrosion.

The MANUFACTURER must request a waiver for any different selection.

6.2 - Equivalent materials

Whenever the MANUFACTURER cannot use the materials specified in paragraph 6.1, he is authorized to propose equivalent materials to TOTAL for approval ; his choice should be substantiated by equivalences recognized by standards and shall be in full accordance with the BS 5500 requirements, including code cases.

Table II

Selection of materials in relation to design temperatures

DESIGN TEMPE-RATURE	+ 427°C to - 196°C	+ 427°C to - 101°C	+ 427°C to - 50°C	MAX 468°C	MAX 538°C	MAX 593°C	Greater than 593°C
TYPE OF STEEL	18/8 stainless steel or 9% Ni	3.5% Ni	Normalized carbon steel	0.5 Mo ---------- 1Cr-0.5Mo	1.25 Cr ---------- 1.25 Cr 0.5 Mo	2.25 Cr-1 Mo ---------- 5 Cr-0.5 Mo	Stainless steel

6.3 - Weldability of C-Mn steels

Carbon - manganese steels shall have a maximum carbon content of 0.23%. The carbon equivalent of these materials, calculated using the formula (IIW) :

$$C_E = C + \frac{Mn}{6} + \frac{Cr + Mo + V}{5} + \frac{Ni + Cu}{15}$$

shall be less than 0.420.

6.4 - Flange gaskets and gasket contact faces

The gaskets are specified in the individual specifications and must comply with the following standards :

- ASME B 16-20 : Ring joint gaskets and grooves for steel pipe flanges.
- ASME B 16-21 : Non-metallic gaskets for pipe flanges.

Except otherwise specified, the following rule shall apply :

Type of gasket	Surface condition of flanges (contact faces) as per MSS SP 6	Mean roughness (maximum value)
ASBESTOS ELASTOMER (series 150 & 300)	1/16 raised concentric serrated or "STOCK FINISH"	150 microns approx. 50 microns
SPIRAL WOUND (series 600)	1/4 raised "SMOOTH FINISH"	250 AARH (or 6.3 microns
RING JOINT (series 900 and above)	SMOOTH (grooved)	63 AARH (or 1.5 microns)

Gasket contact faces for metal joints such as ring joints must be at least 20 Brinell harder than the joints.

6.5 - Bolting

6.5.1 - External bolting

The external bolting of vessels for the assembly of the flanges of manholes, handholes, nozzles with blind flanges, etc... must always consist of threaded studs and nuts. They shall have a protection, against corrosion, of the type AMERI-COTE ("Fluoropolymer Coating").

Threading as per ASME B 1-1 (diameter in inches) :

- UNC for threaded studs up to 1" in diameter,
- 8N for diameters over 1"

6.5.2 - Internal bolting

The internal bolting must comply with this threading. It may consist either of threaded studs and nuts, or of bolts and nuts.

6.6 - Suborders of materials

6.6.1 - Suborders for materials/components shall include ordering information listed in the relevant standard specification.

6.6.2 - Within one (1) week from issue, unpriced copies of all suborders including further change orders shall be issued to TOTAL for information.

7 - RULES OF CONSTRUCTION

7.1 - Standard shell diameter

The outside diameter of the vessel shell shall normally by rounded off to the next 100 mm for standardization. Consideration shall simultaneously be given, however, to the feasibility of pressing heads to the selected diameter at regular cost.

7.2 - Shell and head welded joints number

a) The number of shell longitudinal and circumferential welded joints shall be minimized.

b) Unless otherwise accepted in writing by TOTAL,

- Shell rings 2800 mm or less in outside diameter shall have not more than one (1) longitudinal welded joint.

- The maximum number (n) of circumferential joints, exclusive of head-to-shell joints, shall be as follows :

L	(n)
$\leq$ 2400 mm	0
$\leq$ 4800 mm	1
$\leq$ 7200 mm	2
$\leq$ 9500 mm	3

then one (1) additional joint every 2500 mm of L

Where L = Total length of cylindrical shell between tangent lines.

7.3 - Welded joint layout

Welded joints shall be so arranged that :

a) Visual inspection can be made with all internals in place.

b) No openings intersect with the weld seam.

c) No saddle bearing plates, nozzle reinforcing pads, wear plate..., cover longitudinal or girth joints.

d) No attachment weld edge shall be closer to the nearest weld edge than 1.5 times the wall thickness or 30 mm, whichever is greater.

e) Centers of the longitudinal joints of two (2) adjacent courses shall be staggered 30° minimum apart and preferably be off-centre relative to the vessel main centerlines. For horizontal vessels, they shall be located above the horizontal plane through the centerline of the vessel.

7.4 - Types of welded joints

a) All longitudinal and circumferential joints, and in as much as possible, all pressure retaining joints, shall be double butt-welded.

b) Welds inaccessible for double welding (e.g. circumferential closing joint of small diameter vessel, piping and fitting butt welds...) shall be single butt weld equivalent to a double butt-welded type.

c) Attachment of intermediate heads shall be as shown in the CODE.

7.5 - Heads

a) Heads shall be semi-ellipsoidal, in one piece for carbon, low alloy and clad, steel vessels up to 2300 mm in diameter, with an integral skirt 50 mm long or more.

b) Hemispherical heads may also be used, with TOTAL approval, if not already specified in the vessel data sheet, particularly for high pressure vessels.

c) Other types of formed heads are not permitted.

7.6 - Transitions

Shell-reducer sections shall be toriconical, with a half-apex angle not greater than 30°. Any reinforcement for internal pressure shall be provided by increased plate thickness ; a stiffening ring may be used to satisfy external pressure requirements only.

7.7 - Shell flanges

Full diameter shell flanges may be of the slip-on type only for class up to 300. Over class 300, the welding neck type is required.

a) If the diameter of the vessel is a standardized dimension

The flanges shall comply with one of the following standards :

- ASME B 16-5 up to a diameter of 24" inclusive,
- MSS SP 44 for diameters over 24".
- Alternatively : ASME B 16-30 for the class 150 and 300 (slip-on flanges, regardless of diameter).

b) If the diameter of the vessel is not standardized

The flanges shall be in compliance with standard ASME B 16-30 for the 150 and 300 series (slip-on flanges), or shall be designed in accordance with the BS 5500 code for class 600 and above (Welding Neck flanges).

7.8 - Supports

7.8.1 - Vertical vessels

a) All vertical vessels shall be designed to be free-standing and self supporting on a cylindrical or conical skirt in accordance with TOTAL Standards PS-TCS-208 and PS-TCS-205. The cylindrical skirt plate centerline shall coincide with the shell wall centerline.

b) Vessels 1000 mm OD or less and weighing 1200 kg or less may be supported on structural legs in accordance with TOTAL Standards PS-TCS-209.

c) No brackets, lugs, ring girder nor other type of supports shall be used.

d) Tailing lugs, when required, must be located on the skirt and not on the vessel wall ; they shall be attached to the skirt base ring without interference with the attachment bolts of the vessel. Skirt reinforcement, if needed (e.g. for lifting purpose), must be located outside the skirt envelope. No temporary reinforcements are permitted, wether inside or outside the skirt envelope.

7.8.2 - Horizontal vessels

These vessels are generally supported by two saddles (standard N° PS-TCS-207), located as close as possible to the heads. The MANUFACTURER must calculate the stresses and deformations resulting from these supports and decide whether or not it is necessary to reinforce the vessel.

7.9 - Connections - Nozzles and Manholes

Threaded fittings or couplings shall not be utilized.

7.9.1 - Reference Standards

Connection components shall conform to the following standards :

a) Flanges :

Up to NPS 24 : ASME B 16.5 Pipe Flanges and Flanged Fittings.
Over NPS 24 : MSS SP-44 Steel Pipe Line Flanges.
Flanges shall be of welding neck type.

b) Nozzles :

ASME B 36.10 Welded and Seamless Wrought Steel Pipe.
Carbon and low alloy steel pipes 16 inches and smaller in diameter shall be seamless.

c) Welding fittings :

ASME 16.9 Factory-Made Wrought Steel Buttwelding Fittings.

7.9.2 - Sizes

a) Minimum nozzle size shall be 2" NPS. However, when a connection must be smaller, a reducing flange matching a 2" nozzle neck may be used. Alternatively, a concentric reducer may be inserted between a 2" pipe neck and the required smaller flange. See TOTAL standard PS-TCS-212.

b) Sizes 2 1/2", 3 1/2" and 5" NPS shall not be used.

c) When specified in the individual specification, the inside diameter of nozzles shall be maintained, particularly for connections which must receive equipment or manholes through which internals shall be installed.

7.9.3 - Reinforcement

Added reinforcement plate, when provided, shall have a minimum width of 50 mm and shall be in one piece.

Welded reinforcement should also be avoided when the temperature of the vessel exceeds 400°C, regardless of pressure. In this case, integral reinforcement, tapered for butt-welding to the shell, should be used.

Self-reinforcing forged nozzles shall also be used for high pressure vessels or vessels in Hydrogen service.

7.9.4 - Nozzles for sloping horizontal vessels

When the horizontal centerline of the vessel slopes, the nozzles must be vertical and horizontal and not perpendicular and parallel to the vessel.

7.9.5 - Nozzles for stainless steel vessels

(Walls made of solid stainless steel or covered with stainless steel by cladding, lining or weld deposit).

The nozzles shall be built up as follows :

a) Tubular portion

- For a solid stainless-steel wall : pipe made of the same grade of stainless steel, welded or seamless, according to fabrication possibilities.

- For a carbon or low-alloy steel wall with stainless steel cladding or lining :
  . Diameters up to 3" : solid stainless steel, seamless pipe.
  . Diameters over 3" and up to 16" : seamless carbon or low-alloy steel pipe with stainless steel lining, welded or seamless, according to diameter.

This design requires an expansion ring if the coefficient of expansion of the stainless steel is greater than that of the carbon or low-alloy steel. If the nozzle has a lap-joint flange, the tubular portion is terminated by a forged stainless steel collar for a carbon or low-alloy steel loose flange.

. Diameters over 16" : fabricated tube of carbon or low-alloy steel with stainless-steel strip-lining or cladding.

b) Flanges

Flanges must comply with the following rules :

- Diameter up to 3" : solid stainless-steel flange.

- Diameter over 3" and up to 16" : carbon or low-alloy steel flange with stainless steel weld deposit, or carbon or low-alloy steel lap-joint loose flange on a forged stainless steel collar.

- Diameter over 16" : carbon or low-alloy steel flange with stainless steel weld deposit on the contact face and its standard bore. However, the bores of large-diameter flanges may be covered with stainless steel by strip lining rather than by weld deposit.

c) Thickness of the stainless steel coatings

- The thickness of the lining or cladding of the nozzles must be at least equal to that of the lining or cladding of the shell of the vessel.

- The effective thickness of weld deposit must also meet this condition, with the understanding that the effective thickness means the thickness of the stainless steel coating having the same structure and composition as the stainless steel specified, not counting the dilution zone. In all cases, this weld deposit must be applied in at least two passes.

7.9.6 - Manholes

a) Except when provided with a bolted full diameter end closure, all vessels 800 mm O.D. and larger shall have at least one (1) manhole for inspection, maintenance and operation requirements.

b) Manholes shall conform to TOTAL standards PS-TCS-214, PS-TCS-216 and PS-TCS-218. Alternatively and subject to TOTAL written consent, pad-type manholes may be used when design temperature is 60°C or below and design pressure does not exceed 1 MPa (10 bars).

c) 20" diameter manholes are preferred. Inside diameter of smaller manholes shall not be less than 18 inches (460 mm).

7.9.7 - Connection attachment to shell

a) Typical details are shown in TOTAL standard PS-TCS-213.

b) Connection neck shall be inserted through the vessel wall. Attachment abutting the outside surface of the vessel ("set on") is not permitted.

c) Nozzle and manhole necks shall be attached to the vessel wall and to the separate reinforcement, when added, by full penetration groove welds, preferably double-welded.

d) Except when a piping extension into the vessel is provided, the inner edge of the connection neck shall be flush with the inside contour of the shell and shall have inside corners rounded to 3 mm radius.

e) Connections shall project from the vessel as required for easy access to, and removal of flange studs and nuts, accounting for insulation as appropriate.

f) Bolt holes in flanges of connections located in heads shall straddle the principal centerlines of the vessel, or lines parallel thereto. Bolt holes in flanges of connections located in shell shall straddle the vertical centerline of the vessel.

7.10 - Other attachments to vessel walls

a) Appurtenances welded to the outside surface of the vessel may include stiffeners, lifting lugs, rings, clips and miscellaneous supports for platforms, ladders, davit, piping, insulation, fireproofing or other equipment.

b) Fixed internals may include rings, bars, brackets and clips to support trays and downcomers, piping, gratings and mesh, distributors, etc... as well as baffles, weirs, impingement and wear plates, ladder rungs, etc.

Typical details are shown in the following TOTAL standards :

- PS-TCS-220 : Ladder rungs
- PS-TCS-221 : Vortex breaker
- PS-TCS-222 : Impingement plate
- PS-TCS-225 : Davit
- PS-TCS-229 : Fireproofing supports
- PS-TCS-230 : Insulation supports - Vertical vessels
- PS-TCS-231 : Insulation supports - Horizontal vessels

c) These supports, except as otherwise indicated in the individual specification, must be made of the same grades of material as the walls to which they are welded.

d) Any interference between these supports and the shell welds or between the supports themselves must be reported by the MANUFACTURER to the ENGINEERING COMPANY before the supports in question are set in place.

e) Welds :

- Attachments welds to the vessel wall shall be designed to the same allowable stress limits required for pressure retaining parts.

- Attachments shall be continuously welded to the vessel.

- All doubler plate pads welded to the vessel shall have at least one (1) 6 mm telltale hole in lower position of pad.

7.11 - Lifting devices

Vessels weighing fifteen (15) tonnes or more shall have lifting lugs or trunnions designed with a safety factor or four (4) from the minimum specified ultimate tensile strength (U.T.S.) of the material. When required by the individual specification, vessels less than fifteen (15) tonnes shall also be provided with lifting lugs.

See TOTAL standard PS-TCS-204 for vertical vessels.

7.12 - Internals

Removable internals shall be designed to pass through the vessel manhole(s) and shall allow easy installation and removal.

Internal piping 6" NPS and larger shall be flanged for ease of removal.

8 - FABRICATION

8.1 - General

a) Fabrication shall not commence until the MANUFACTURER has received written authorization from TOTAL.

b) The MANUFACTURER shall not subcontract any work without previous written acceptance by TOTAL. This acceptance may require from TOTAL an audit at the sub-contractor premises.

c) Machined faces and edges shall be protected against damage during fabrication and against oxidization during heat treatment.

8.2 - Cutting

Shear cutting shall not be utilized for pressure retaining plate material.

8.3 - Forming

a) Plate forming shall normally be parallel to the final direction of mill rolling.

b) With the exception of heads, forming above 150°C shall not be made unless MANUFACTURER's detailed procedure is previously accepted in writing by TOTAL.

c) Cold forming by "step rolling" or bending of steels having a Yield Strength over Ultimate Tensile Strength ratio larger than 0.75 is not permitted. This process is also not permitted on Quenched and Tempered steels.

8.4 - Welding

Welding shall comply with the TOTAL General Specification SP-TCS-611 : "Welding of pressure containing piping and equipment", and the following additional requirements :

8.4.1 - Welding technique

a) Whenever practicable, welding shall be done in the flat position.

b) Welds made of only one pass are not acceptable on pressure containment welds.

c) Gas metal arc welding (GMAW) is not acceptable, even on root passes.

d) All double-sided butt-welds shall be back-gouged by arc-air followed by grinding down to sound metal prior to back-welding.

8.4.2 - Welding documents

Before starting any fabrication work, the MANUFACTURER must submit one set of welding documents per vessel, including at least the following items of information, to the ENGINEERING COMPANY and the INSPECTION AGENCY for approval :

a) Reference :

Order number of TOTAL or ENGINEERING COMPANY.
Name and identification of the vessel.
Place of fabrication.

b) Weld map :

A sketch of the vessel showing the types of joint and seam locations for the vessel and such welded fittings as supports, gussets, reinforcements, internals, etc..., and mentioning the dimensions of the plates (length, width, thickness) and of the pipes (diameter, thickness, length). Also :

- grades of materials,
- joint efficiencies,
- construction and inspection code.

Each weld of this weld map must be cross-referenced to a WPS number.

d) Welding Procedure Specification (WPS) including :

- The identification number of the qualification test record supporting each welding procedure specification (record to be attached).
- Identification of welds.
- Type of preparation (oxy-gas cutting, grinding, machining, etc...).
- Welding process.
- Welding position.
- Identification of welder(s) or operator(s).
- Filler material (wires, electrodes) and diameters.
- Protecting flux.
- Refractory (tungsten) electrode (diameter).
- Characteristics of electric current.
- Speed of execution of each pass.
- Preheating, post-heating.
- Welding equipment.
- Gouging (type).
- Post weld heat treatment.
- Examinations : radiography, ultra-sonic, dye penetrant, etc...

8.4.3 - Approval of welders

The MANUFACTURER must issue a list of approved welders. This list must specify, in particular :

- The identification number of each welder.
- The number and date of the certificate of qualification (certificate to be attached).
- The term of validity.

This list must be certified by the INSPECTION AGENCY.

8.5 - Post-weld heat treatment

8.5.1 - General

A post-weld heat treatment must be performed if it is required by the Basic Code and/or local regulation.

The heat treatment record chart must submitted by the MANUFACTURER (paragraph. 13) to the INSPECTION AGENCY's approval.

8.5.2 - Specific requirements

A post-weld heat treatment must also be applied in the following cases :

a) Vessels made of Carbon Steel materials and operating in a "sour wet" condition, as defined by NACE MR 01-75 ("Sulfide stress cracking resistant metallic materials for oil field equipment").

b) Vessels containing alkaline solutions must undergo postweld heat treatment when the design temperatures are higher than the following temperatures :

$$T = 73 - \frac{25}{46} N$$

where : T = limiting temperature without treatment (°C)
N = content in percent (by weight) of caustic soda in the solution.

The treatment is not necessary for solutions :

- of which the concentration by weight is less than 1.5% ;
- at ambient temperature, regardless of concentration.

c) Vessels containing solutions of potassium carbonate at concentrations over 2% (by weight) must be post-weld heat treated when the design temperatures are over 90°C or when the vessels are equipped with reheaters without temperature alarms.

d) Vessels containing solutions of monoethanolamine must undergo post-weld heat treatment.

e) The MANUFACTURER is authorized to propose a heat treatment when he deems it necessary, even if neither the Basic Code nor Local Regulations require it.

f) Other cases may require heat treatment ; this is then stipulated in the individual specifications.

8.6 - Production test pieces

Production test pieces shall be provided in the following cases :

a) When the basic code or local regulations require it.

b) When the vessel must undergo one of the heat treatments provided for in paragraph 8.5.

For each vessel to which one of the above cases applies, the MANUFACTURER must prepare two test pieces using the same welding procedures as those used for the assembly of the vessel and the same materials (basic metal and filler metals), at the same time. The thicknesses of the test pieces must match the thickness of the thickest wall, not including any reinforcements. These test pieces must furthermore undergo the same heat treatments as the vessel itself, in the same furnace and at the same time.

The mechanical tests to be carried out on these test pieces shall be those required for type approval of the corresponding weld procedures, and their results must satisfy the same criteria.

Mechanical tests shall be carried-out on only one of the two test pieces, the second one being kept for any necessary investigation or re-test in the future. This second coupon shall not be cut or touched in any way without the prior acceptance of TOTAL.

The results must be certified by the INSPECTION AGENCY and reported to the ENGINEERING COMPANY.

9 - PRESSURE TESTS

Each vessel must undergo a pressure test in compliance with the Basic Code.

This test must be made at the pressure determined according to this code.

The MANUFACTURER shall submit the test pressure calculations to the ENGINEERING COMPANY for approval.

When the vessel is subject to local regulations, it may have to undergo a second official test following the first one.

The tests must show that the vessels are perfectly tight and that there are no permanent deformations.

9.1 - Shop tests

Without exception, the shop tests are hydraulic tests (i.e. with water). However, the ENGINEERING COMPANY may propose pneumatic tests to the MANUFACTURER in certain special cases. The MANUFACTURER must agree or comment before the order is placed.

Transportable vessels undergo the above-mentioned tests in the shop, in a position compatible with their dimensions. However, for vessels that must undergo retesting in their operating positions, the shop test pressures must be increased by the corresponding hydrostatic heads.

9.2 - Field tests

Field-assembled vessels are pressure tested in their final positions.

Certain vessels operating under gas pressure are tested pneumatically when the foundations make it impossible to fill them with water. In this cases, test procedures must be drawn up by the MANUFACTURER and submitted to the ENGINEERING COMPANY and to the LOCAL AUTHORITIES for comments.

9.3 - Testing procedure

a) The vessels shall be tested in the presence of the INSPECTION AGENCY, before being painted.

b) Prior to testing, the vessels shall be thoroughly cleaned and free from dirt, debris, loose scale and slag, pieces of metal, weld spatter, oil and grease, etc...

c) Tightness of welded attachments with telltale hole shall preliminarily be air and soapsud tested.

d) Service bolting shall not be used for nozzles and manholes.

e) Gaskets shall be of the service type, dry or coated with graphite. Use of compounds, glue, lead... is not permitted.

Damaged gaskets shall be replaced after testing.

f) All air shall be vented from the vessel before the pressure is applied.

g) Test pressure shall be held at least one (1) hour during visual examination of the vessel by the INSPECTION AGENCY.

h) Immediately after draining, vessel shall be dried thoroughly.

i) The MANUFACTURER shall supply blanks and bolting to be re-used for further site testing of the vessels.

9.4 - Temperature and quality of the test water

Testing medium shall be clean, fresh water. When testing stainless steel vessels, the water shall have a maximum chloride content of 15 ppm and shall be inhibited ; the inhibitor shall be constituted of 3 ppm of sodium nitrate (Na N03) per ppm of chloride in water, and of 12 ppm of oxygen scavenger per ppm of oxygen in water.

If vessel envelope materials (other than austenitic stainless steels) are not impact tested, both the test water and the envelope metal temperatures shall be a minimum of + 16°C. In all other cases, these temperatures shall be at least 10°C above the impact test temperature of enveloppe materials, with a minimum of + 10°C.

10 - INSPECTION AND QUALITY ASSURANCE

10.1 - General

Materials, design, fabrication and preparation for shipment of each vessel shall be subject to inspection by TOTAL and/or the INSPECTION AGENCY designated in the contract and, if applicable, by the certification authority.

However, the acceptance of any work and/or release of the vessels by TOTAL and/or the INSPECTION AGENCY shall in no way relieve the MANUFACTURER of any responsibility for carrying out all the provisions of the CODE and this specification.

Unless otherwise stated in the contract, all qualifications, examinations, tests or inspections shall be made at the place of fabrication.

The MANUFACTURER shall notify TOTAL and the INSPECTION AGENCY, at least ten (10) working days in advance, of the date of tests or inspections to be witnessed.

10.2 - Quality Assurance system

For confidential evaluation and audit, TOTAL and/or the INSPECTION AGENCY may require the written description of the MANUFACTURER's Quality Assurance System.

10.3 - Quality plan

Within six (6) weeks from the effective date of contract, the MANUFACTURER shall set out a step by step inspection program for TOTAL and the INSPECTION AGENCY approval.

This program shall be in general accordance with the provisions of the CODE and shall itemize all required stages of inspection with hold points.

10.4 - Material verification, inspection, and testing

10.4.1 - General

Unless otherwise stated in the individual specification, testing shall conform to the requirements of the applicable standard(s).

Plates 30 mm and thicker shall be ultrasonically examined in accordance with ASME Code Section V and SA 435 (periphery and grid pattern).

Other suitable supplementary testing or examination may include, but are not limited to :

- Chemical analysis (product analysis)
- Grain size
- Etching test
- Mechanical testing :
  - . after simulated postweld heat treatment (if applicable)
  - . additional tensile tests :
    - * at design temperature
    - * on the opposite end of each plate-as-rolled

. reduction of area
. bend test
. flattening test
. hardness test
. impact test

- Corrosion test
- Ultrasonic testing, magnetic particle, liquid penetrant examinations.

10.4.2 - Repairs

No repairs shall be made unless previously accepted in writing by TOTAL.

10.4.3 - Material test reports

For all pressure parts (plates, heads, nozzles, flanges, etc...) as well as for supports (skirt, saddles, legs), the MANUFACTURER shall provide material test reports showing the results of the required tests and inspection, including supplementary requirements if any. Material certificates shall conform to DIN 3.1.B.

Certificates of compliance to product specification may be accepted for non pressure part materials only.

10.4.4 - Material identification sketch

For each vessel, the MANUFACTURER shall prepare a material identification sketch showing the marking on every piece and referencing the related material test report or certificate.

10.4.5 - Transfers of markings

Transfers of markings shall be made by the MANUFACTURER prior to cutting and shall be witnessed by the INSPECTION AGENCY.

10.5 - Welding inspection

10.5.1 - Visual and nondestructive examinations

Visual and nondestructive examinations of weld preparation and welded joints shall conform to the requirements of the Basic Code, the General Specification TOTAL N° SP-TCS-611 and the following :

a) Nondestructive examination procedures shall be submitted for approval to TOTAL and the INSPECTION AGENCY before fabrication starts.

b) Spot radiography shall include all butt-weld intersections and not less than ten (10%) percent of each longitudinal and circumferential joint length in the shell and heads.

c) The INSPECTION AGENCY shall designate the locations at which spot radiographs will be taken.

d) Nozzle attachment welds to shell shall be 100% inspected by ultrasonic examination.

e) In addition to the requirements of the General Specification SP-TCS-611, attachments to pressure parts including intermediate heads, lifting lugs and skirt-to-shell weldments shall be subject to magnetic particle or liquid penetrant examination.

   Fillet and butt welds which cannot be radiographed or ultrasonically examined shall be checked by magnetic particle or dye penetrant examination in accordance with the method specified by the Basic Code.

   This requirement applies to all welds, including the welded zones of temporary fabrication attachments.

f) All butt-welds of high pressure vessels shall be 100% radiographed and 100% Ultrasonically tested.

g) All NDT including radiographic, ultrasonic, magnetic particle or liquid penetrant examinations shall be performed before and after Post Weld Heat Treatment of the Vessel.

h) Radiographic inspection must be done by X-rays. However, radiography may be performed using gamma rays only if AGFA GEVAERT D4 or KODAK M film quality is used, and subject to prior written approval of TOTAL.

i) Qualification of NDT personnel :

   - all operators, including for Dye Penetrant or Magnetic Particle Inspection, must be qualified to SNT-TC-1A level I, as a minimum,
   - interpretation must be carried-out by inspectors holding at least a Valid Certificat to SNT-TC-1A level II or CSWIP.

10.5.2 - Hardness

Brinell hardness of base metals, welds and heat affected zones shall not exceed 225 for base materials containing a minimum of 0.5% chromium or 0.5% molybdenum.

10.5.3 - Dimensional check

The final diemnsional check and visual inspection shall be performed after the hydrostatic test. Tolerances shall be as shown in TOTAL standard n° PS-TCS-201.

If so required in the individual specification, removable internals shall be trial fitted inside the vessel.

11 - PREPARATION FOR SHIPMENT

11.1 - General

No vessel or vessel section shall be released for shipment without TOTAL prior written authorization.

11.2 - Particular requirements

a) After drying out, the vessels or sections shall be cleaned externally and internally of all foreign matter, dirt, rust, scale, grease...

b) Internal surfaces of stainless steel vessels shall be pickled and passivated.

c) External surfaces of carbon steel vessels shall be blasted and painted as prescribed in the individual specification.

d) Orientation of vertical vessels shall be clearly marked with white paint on the top and bottom portions of the shell n° 0°, 90°, 180° and 270° respectively.

e) All machined surfaces shall be cleaned with a suitable solvent and immediately be protected with an easily removable rust preventive coating or plastic film.

Surfaces adjacent to gasket faces shall be protected in the same manner for a distance of 100 mm.

f) All connection openings shall be sealed with one piece plywood covers or heavy plastic caps taped to the nozzle.

Permanent blind flanges or covers included in the MANUFACTURER's supply shall be bolted with gaskets, stud bolts and nuts in place, ready for service.

g) Telltale holes shall be fitted with wooden plugs or may be packed with sealant grease.

h) Internals installed prior to shipment when required shall be properly secured and protected.

i) Removable internals, davit, anchor bolts, spares and all other loose items as required in the individual specification, shall be adequately marked or tagged for identification and packed in fully waterproof and moisture proof lined wooden crates, items being separately wrapped in a plastic sheet, if appropriate.

Copy of the packing list shall be fixed inside the crates.

11.3 - Nameplate

Each complete vessel shall be provided with a nameplate as shown in TOTAL standard PS-TCS-223, securely attached and so located that it is easily accessible after the vessel installation. The nameplate shall be on a bracket welded to the shell. Insulated vessels shall have bracket to provide 25 mm clearance between nameplate and the outside surface of insulation.

12 - FITTINGS AND SPARE PARTS

12.1 - Fittings

All fittings mentioned in the individual specifications are included in the supply. These include, in particular :

- Manhole covers with their handling apparatus (davit).
- Major supports (saddles, skirts, legs).
- Other supports (for insulation, structures, platforms, piping, etc...).
- Blind flanges for tests and the corresponding studs and gaskets.
- All internals together with their attachment bolts and other fittings.

12.2 - Spare parts

The spare parts shall include at least :

- 2 sets of gaskets (including manholes gaskets)
- 5% of studs, bolts, nuts, washers, etc... of each type (with a minimum of 2 units)
- other parts, such as internals, if required in the individual specifications.

13 - DOCUMENTATION

13.1 - a) The MANUFACTURER's documentation shall be :

- identification as required in the contract,
- in English language,
- in S.I. units (except nozzles and flanges which shall be in nominal inch size).

b) Sizes of drawings shall be A1 or A2.

13.2 - For each vessel, the MANUFACTURER shall submit for TOTAL review printed and/or reproducible copies of engineering design and fabrication documents and subsequent revisions including :

a) Within three (3) weeks from the effective date of contract, complete design analyses and calculations sheets or computer printouts, as well as the quality plan.

b) Within four (4) weeks from effective date of contract, the shell and head layout drawing showing the location and orientation of the longitudinal and circumferential welded joints.

c) Within eight (8) weeks from the effective date of contract, all the fabrication drawings :

1 - Outline and assembly drawing, incorporating design basis and vessel data :

- Applicable code
- Applicable TOTAL specifications
- Maximum service pressure (M.S.P.)
- Maximum and minimum service temperatures (M.S.T. and m.S.T.)
- Design pressure(s)
- Design temperature(s)
- Other loadings
- Corrosion allowance
- Joint efficiency
- Insulation
- Materials specifications
- Post Weld Heat Treatment (P.W.H.T.)
  (Procedure n° ..... date .....)
- Maximum allowable working pressure (M.A.W.P.)
  Limited by .....
- Hydrostatic test pressure
- Lifting weight
- Operating weight
- Hydrostatic test weight

2 - Detail drawings :

- Nozzles
- Supports
- Internals
- Welding details

3 - Welding doucments :

- Weld map
- Welding procedures specifications (WPS)
- Procedures qualification records (PQR)

4 - Fabrication procedures :

- Post weld heat treatment procedure
- NDT procedures
- Pressure test procedures